Xiang LI

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EDUCATION BACKGROUND

Nanjing Institute of Technology (NJIT) ***2022.09-2026.06***
- **Degree:** Bachelor of Engineering in Software Engineering **GPA:** 85.08/100

University of Oulu ***2022.09-2026.06***
- **Degree:** Bachelor of Science in Software Engineering **GPA:** 4.04/5.0

City University of Hong Kong ***2026.08-2027.06***
- **Degree:** Master of Computer and Information Engineering
- **English:** IELTS (6.5) **Computer:** proficient in MySQL, MongoDB, Linux, Android, Windows, C language, C++, Java, SQL, HTML/CSS, JavaScript, and Python, AI Agent

RESEARCH & COMPETITION

First Author & Corresponding Author, **UAV Emergency Image Classification Based on ConvNeXt and Swin Transformer: A Comparative Study on Optimization Strategies and Performance** ***2025.06-2025.08***
- Identified the limitations of current drone disaster recognition: small models lacked accuracy and large models ran too slowly, compared CNN and Transformer models to find the most suitable one.
- Split the dataset based on a 7:1:2 ratio and tested both CNN and Transformer models, ultimately selecting ConvNeXt-small and Swin Transformer as benchmark models.
- Fine-tuned the selected benchmark models using the AdamW optimizer and experimented with Layer-wise Learning Rate Decay (LLRD) strategy, compared the effects of data preprocessing methods such as CutMix+MixUp, RandAugment, and RandomErasing, and validated the effectiveness of attention mechanisms.
- Determined the combination of Swin Transformer model with AdamW optimizer, LLRD strategy, and specific data augmentation methods delivered the best performance, achieving higher accuracy and a global layout understanding.
- Wrote a thesis published on AIAHPC 2025, will be published on IEEE(ISBN: 979-8-3503-9237-1), will be submitted for index in IEEE Xplore, EI Compendex, Scopus.

Algorithm Implementer, **Optimization and Dynamic Prediction of Human Pose Estimation Based on Kalman Filter and Visual Detection** ***2025.01-2025.03***
- Extracted the key features of human body based on the principle of Kalman filter algorithm and visual detection, leveraged Python, OpenCV and deep learning frameworks to realize the visual detection module; applied the Kalman filter for real-time tracking and smoothing of key point trajectories to optimize pose recognition performance.
- Collected and sorted out various posture datasets; preprocessed data, including data cleaning, normalization, and augmentation, to enhance the model's generalization capability; adjusted Kalman filter parameters to improve the algorithm's accuracy and real-time responsiveness in dynamic scenarios.
- Integrated the algorithm into the human pose estimation system, tested it in multiple scenarios, optimized and achieved the algorithm able to improve the accuracy, real-time responsiveness, and dynamic estimation of human pose.
- Published the outcome in the journal *Practical Electronics*.

Member, **Jiangsu Province University Innovation Training Program: Research on the Privacy Protection of Energy Internet Based on Differential Privacy** ***2023.01-2024.12***
- Aimed to combine differential privacy and intelligent reflecting surface (IRS) technology to optimize privacy protection mechanisms in energy internet and enhance energy transmission efficiency
- Participated in designing a differential privacy protection model suitable for energy internet scenarios, and optimized the application of the Laplace mechanism and the exponential mechanism.
- Engaged in the research and integration of IRS technology, assisted in designing IRS deployment schemes, analyzed its optimization role in energy transmission, and conducted testing and verification.
- Employed non-intrusive load monitoring (NILM) technology, explored its performance in privacy protection and data accuracy, proposed optimization suggestions and helped with system integration.
- Contributed to integrating differential privacy algorithms, IRS and NILM technology, built a privacy protection framework, validated its performance in different experimental scenarios, and wrote a report.

Member, **China Robot Competition and RoboCup Robot World Cup China (Third Prize)** ***2024.08***
- Intended to achieve the robot's functions of autonomous obstacle avoidance and precise targets attacking in dynamic environments through the collaboration of visual and decision systems.
- Developed a multimodal perception system framework based on Python and ROS, integrated the speech recognition API of iFLYTEK to trigger instructions; generated the obstacle avoidance path planning by DWA, YOLOv5 and LIDAR.
- Used the DNN module from OpenCV to load a pre-trained FaceNet model for fast face detection and feature extraction; implemented identity recognition by Euclidean distance comparison, minimized computational overhead while ensuring basic functionality; kept processing latency on the Jetson Nano platform under 200ms to meet real-time requirements.
- Optimized module response delay and parameter settings on a simulated track, adjusted face recognition thresholds under varying lighting conditions, and refined obstacle avoidance algorithms on a basis of obstacle distribution.
- Achieved a 90% task completion rate, 95% obstacle avoidance success rate, and 95% facial recognition accuracy.

Participant, **Lanqiao Cup Competition** ***2024.04***
- Applied the Depth-First Search (DFS) algorithm to tackle graph theory problems, explored the search tree to identify the optimal solution, and applied pruning techniques to enhance algorithm efficiency.
- Employed the Dynamic Programming (DP) algorithm to resolve the maximum sum finding problem, leveraging the state transition equation, optimal substructure properties, and memoization to achieve efficient computation.
- Optimized the shortest path finding problem using the Breadth-First Search (BFS) algorithm, implemented level-order traversal with a queue data structure to quickly locate the shortest path, and took into account the non-negative nature of edge weights.
- Awarded with the **First Prize** of Java Competition in Jiangsu Provincial Zone (Top 10%), and the **Third Prize** in National Finals.

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INTERNSHIP

Software Development Intern, **GSC China HD&VF-SSVF-CS&Q, Schneider Electric** ***2026.03-2026.07***

- Independently designed an intelligent supply chain data governance platform, upgrading the traditional statistics system into a multi-agent collaborative system based on LangGraph4j, covering the full lifecycle management of 200,000 supplier records per week.
- Defined the agent execution topology using LangGraph4j StateGraph, modeling scheduling, validation, prediction, root-cause attribution, and notification as independent Nodes; declaratively orchestrated subtask dependencies (import → validation → prediction → leaderboard refresh → notification) via Conditional Edges; leveraged LangGraph4j's asynchronous node execution (CompletableFuture) and parallel edges to achieve fan-out parallel dispatch and result aggregation, compressing the end-to-end processing pipeline to < 15 seconds.
- Implemented a prediction anomaly auto-attribution Agent that retrieves similar scenarios from a vector database and generates attribution reports when deviation exceeds the threshold; built a notification Agent (WebSocket + Email) for targeted push alerts with role-based customizable strategies.
- Based on Spring Boot + MyBatisPlus + MySQL, achieved < 5 second writes for 200,000 records through streaming reads and batch writes; optimized composite indexes to deliver millisecond-level leaderboard query responses; enabled one-click deployment via Docker containerization.
- Platform impact: anomaly detection time reduced by 70%, prediction coverage reached 90% of management meetings, and employee data confirmation efficiency improved by 40%.

Software Development Intern, **Google Information Technology (Shanghai) Co., Ltd.** ***2024.12-2025.03***

- **Satellite Image Annotation Project:**
  - Investigated the market to identify user needs: viewing satellite image markers on a map, clicking markers to view thumbnail previews, and downloading high-resolution images.
  - Utilized JavaScript on the frontend and integrated the Google Maps API for map rendering and interaction.
  - Employed Java on the backend, and applied the Spring Boot framework to handle map data and user requests.
  - Used a MySQL database to store annotation data and user information, while incorporating Redis as a cache layer to store hot data and enhance system performance.
  - Completed performance testing and deployed the system.
- **Development of an E-commerce Management System Based on the Spring Boot Framework:**
  - Determined functional requirements: modules such as product display, order management, and user management.
  - Adopted Vue.js to build product list and detail pages, supporting image upload and rich text editing.
  - Implemented CRUD operations for product information via Spring MVC and MyBatis.
  - Stored system data in a MySQL database and leveraged Redis to cache popular product information.
  - Completed module development, integrated payment interfaces, and delivered the e-commerce management system.
- Assisted in developing other software, compiled and optimized test case libraries, and boosted the stability of software.
- Monitored the operating system environment, resolved problems, and extended its mean time between failures.

Software Development Intern, **Hangzhou Bijisoft Co., Ltd.** ***2024.07-2024.08***

- Delved into users' requirements for the software products, clarified development goals and product functionalities, and developed modules via Java to accomplish functional code implementation.
- Collaborated in code reviews and unit testing to make sure code quality and functional accuracy; drafted technical documentation.
- Participated in integration and system testing, tackled challenges identified during testing to guarantee system stability.
- Adjusted and optimized the software according to user feedback, thereby enhancing product user experience and stability.

EXCHANGE EXPERIENCE

University of Cambridge (ONLINE PROJECT-BASED LEARNING PROGRAM) ***2025.01***

- Attended modules like *Introduction to Robotics*, *Computer Vision*, *Artificial Intelligence* and *Smart Healthcare.*
- Explored robotic hand designs and their applications; constructed computer vision models; developed reinforcement learning models; dived into the applications of AI in smart cities, autonomous vehicles, and smart healthcare.
- Designed a human pose prediction system, and published it on a journal.
- Achieved an A-level grade in the online course *Artificial Intelligence and Robotics*.

PUBLICATION & PATENT

First Author & Corresponding Author, UAV Emergency Image Classification Based on ConvNeXt and Swin Transformer: A Comparative Study on Optimization Strategies and Performance, *AIAHPC 2025*, 225072814412197419
Third Author & Corresponding Author, Optimization and Dynamic Prediction of Human Pose Estimation Based on Kalman Filter and Visual Detection, *Practical Electronics,* accepted, 2025.09
Steering Wheel All-terrain Chassis, Utility Model Patent, No.: 202520729515.9, 2025.04

HONORS & AWARDS

Second-class Scholarship, NJIT ***2025.03; 2023.02***
Third-class Scholarship, NJIT ***2023.02; 2022.09***
Merit Student ***2023.02***
National Third Prize, "Huashu Cup" Contemporary Undergraduate Mathematical Contest in Modeling ***2025.08***
Jiangsu Provincial Third Prize, Group Programming Design Competition ***2024.04***
Excellence Award, 5th China Collegiate Algorithm Design & Programming Challenge Contest ***2024.03***
Second Prize, Jiangsu Provincial Practical Contest - Cloud Service, Huawei ICT Competition ***2023.04***